Multimodal Remote Sensing Image Registration via Modality Perception and Self-Supervised Position Estimation

Yun Xiao, Chunlei Zhang, Bo Jiang, Yuan Chen, *Member, IEEE*, and Jin Tang

***Abstract*— Multimodal remote sensing image registration ensures that images from different sensors or modalities are spatial and informational consistent for effective comparison and analysis. However, due to the nonlinear modality gaps that exist between images, it is difficult to focus solely on spatial positional differences while ignoring the modality gaps. In this article, to address this issue, we propose a new framework for multimodal registration network, named MMRNet. The proposed framework comprises the following main aspects. First, a novel self-supervised positional misalignment estimator (PME) is designed for multimodal image registration. PME can efficiently overcome the modality gaps and learn the positional differences between multimodal images more reliably, optimizing the registration loss by minimizing the positional differences directly. Then, a new paradigm of modality translation, termed modality perception module (MPM), is introduced to effectively learn modality gaps and perform modality translation in the case of positional misalignment. Finally, we further design the modality perception guidance loss to supervise the modality translation task, which can encourage the fidelity of the generated pseudo-modality images. Our registration network integrates both rigid registration model and nonrigid registration model. The experimental results demonstrate that the proposed registration framework can obtain obviously superior performance in both rigid and nonrigid image registration tasks on optical-synthetic aperture radar (SAR) data, optical-map data, and optical-infrared data. The code and relevant dataset will be made publicly available at https://github.com/Ahuer-Lei/MMRNet.**

***Index Terms*— Image registration, modality translation, multimodal learning, remote sensing processing, rigid and nonrigid transformation.**

I. Introduction

MULTIMODAL remote sensing images exhibit unique perceptual advantages, such as optical image boast high resolution and rich color information, synthetic aperture radar (SAR) images offer all-weather imaging capability, and infrared images reflect the thermal characteristics of the target. Aligning them in the same coordinate system enables comprehensive observation and analysis, resulting in significant implications in practical applications. However, nonlinear modality gaps between multimodal images due to different imaging mechanisms, such as the inevitable speckle noise in SAR images, bring several challenges in the registration process.

In recent years, multimodal learning has received increasing attention due to its ability to integrate complementary information from heterogeneous sources. Several advanced frameworks have been proposed for classification and semantic understanding tasks. For instance, Wu et al. [1] explore convolutional neural networks (CNNs) for multimodal remote sensing data classification, revealing the potential of deep feature fusion. More recently, Li et al. [2] propose FlexiMo, a flexible remote sensing foundation model designed to tackle diverse multimodal tasks. These works underscore the growing demand for general and robust multimodal methods in the remote sensing field. However, compared to these tasks, multimodal image registration remains relatively under-explored. As a prerequisite for multimodal image fusion and analysis, image registration plays a vital role in applications such as object recognition and change detection. Therefore, developing a general and robust multimodal image registration approach is essential for enhancing the comprehensive utility of remote sensing imagery.

In the past two decades, with advancements in computer vision and deep learning networks, there has been significant research interest in learning-based multimodal image registration methods [3], [4], [5], [6], which can be broadly categorized into two types. One type is the unsupervised learning, where a common method is a similarity-based registration. This method utilizes image similarity operators, such as NCC [7], [8], [9], MI [10], [11], [12], and MIND [13], as the loss function for registration. However, these operators have inherent limitations because they are designed based on prior mathematical knowledge. First, designing a generalized similarity operator for all data is challenging. Second, estimating the upper bound of the performance of these similarity operators is impossible, and thus, finding directions for improvement is difficult [14]. The other is supervised learning, which optimizes the network by minimizing the difference between predicted and ground-truth transformation parameters.

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Yun Xiao and Chunlei Zhang are with the Key Laboratory of Intelligent Computing and Signal Processing, Anhui Provincial Key Laboratory of Security Artificial Intelligence, School of Artificial Intelligence, Anhui University, Hefei 230601, China (e-mail: xiaoyun@ahu.edu.cn; ahdxyjszcl@163.com).

Bo Jiang and Jin Tang are with the Information Materials and Intelligent Sensing Laboratory of Anhui Province, Anhui Provincial Key Laboratory of Multimodal Cognitive Computation, School of Computer Science and Technology, Anhui University, Hefei 230601, China (e-mail: jiangbo@ahu.edu.cn; tangjin@ahu.edu.cn).

Yuan Chen is with the School of Internet, Anhui University, Hefei 230601, China (e-mail: ychen@ahu.edu.cn).

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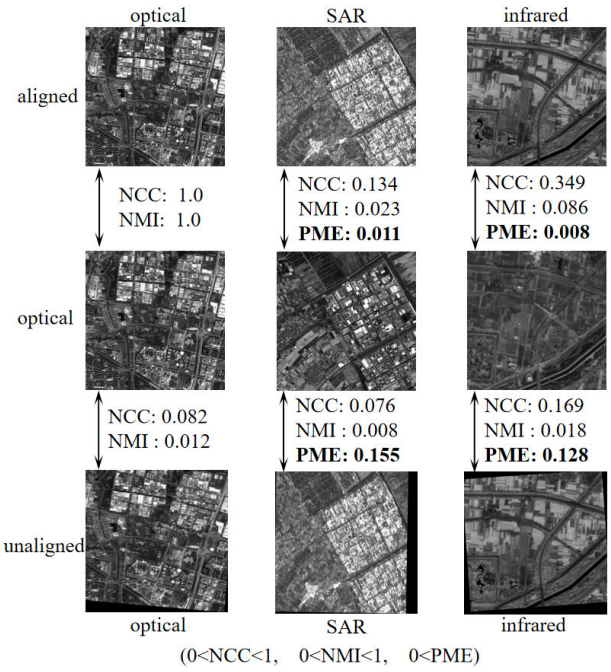


Fig. 1. Larger values of NCC and NMI indicate that the image pairs are more similar, and smaller values of PME indicate that the image pairs have less positional error. (First column) NCC and NMI can effectively measure the accuracy of registration between optical image pairs. (Second and third columns) When NCC and NMI are used for multimodal images, the similarity computation results in lower values even for fully aligned image pairs. Our PME can differentially evaluate the degree of registration.

Many existing supervised methods [15], [16], [17], [18] typically require the ground-truth transformation parameters between images by artificially distorting the images to misalign them. This strategy is effective for mono-modal image registration, but when applied to cross-modal images, the learned parameters become unreliable due to significant modality gaps. Therefore, seeking some richer and more effective supervisory information is desirable to optimize the registration loss. Most of the methods use similarity operators such as NCC and normalized mutual information (NMI) as auxiliary objectives to enforce constraints from semantic information, which constitutes a dual-supervised loss. But as mentioned above, these losses cannot universally adapt to the general imaging modalities, as shown in Fig. 1. Some novel similarity metrics have been developed to address this task [19], [20]. In addition, some other studies [21], [22], [23], [24], [25] address the modality gaps in cross-modal registration by employing modality translation. The specific approach involves using generative adversarial networks (GANs) [26] to translate cross-modal images into mono-modal images. However, training the GANs is inherently challenging and may generate additional artificial noise that makes the registration process more difficult [27]. Meanwhile, GAN-based approaches cannot guarantee consistent instance correspondences between the translated and target images due to their principle of remapping data distributions between modalities.

To address the above issues, in this article, we propose a novel general approach for the registration of multimodal remote sensing images, termed multimodal registration network (MMRNet). Overall, there are three main aspects of the proposed approach. First, to explore effective supervisory information, inspired by [14], we devise a self-supervised evaluation module, termed positional misalignment estimator (PME). From a self-supervised perspective, our introduced PME effectively mitigates the influence of modality gaps to accurately estimate positional differences between the two images. Our PME is implemented based on a simple ResNet [28] architecture, endowing it with robust learning capabilities and broad applicability. Second, to alleviate registration challenges, we introduce a new modality perception module (MPM) for modality translation. Different from GAN-based methods, our MPM directly learns modality differences between images, ensuring a reliable similarity metric between the translated image and the target image. Finally, to encourage the stable and effective operation of MPM, we further design a modality perception guidance loss to guide the whole learning process. In conclusion, MMRNet overcomes the limitations of traditional similarity operators by replacing them with PME, which significantly enhances both registration accuracy and robustness. Additionally, the introduction of MPM effectively reduces the impact of modality gaps. By integrating two complementary registration models, MMRNet can handle both rigid and nonrigid registration tasks, providing a unified and adaptable framework. As a result, our method establishes a unified registration framework with strong generality and improved generalization capability.

Moreover, due to the absence of publicly available datasets, we have created a new dataset for multimodal remote sensing image registration, named the MRSR dataset. Specifically, the proposed dataset includes optical-SAR data and optical-infrared data where the former contains 3850 pairs of optical-SAR images of size 256×256 with a resolution of 10 m, while the latter contains 4000 optical-infrared pairs of the same quality. The following factors are taken into account to make our data more diverse, including different seasons and scenes.

In summary, the main contributions of this article can be summarized as follows.

1) We propose MMRNet, a unified registration framework capable of handling both rigid and nonrigid transformations across diverse modality pairs. Unlike prior works that typically focus on either rigid or nonrigid settings, MMRNet offers a flexible and generalizable solution for multimodal image registration tasks.
2) We construct MRSR, a new benchmark dataset for multimodal registration, comprising optical-SAR and optical-infrared image pairs.
3) We design a PME, which provides more effective supervision than traditional similarity operators by learning directly from data, thereby avoiding the limitations of handcrafted metrics and improving adaptability across modalities.
4) We introduce an MPM along with a modality perception guidance loss, enabling the network to learn modality-specific discrepancies while maintaining consistent spatial correspondence.

The experimental results demonstrate that the proposed method can achieve superior performance in multimodal

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remote sensing image registration tasks when compared to recent registration methods.

The remainder of this article is structured as follows. Section II provides an introduction to the related work. Section III offers a detailed description of our methods. Section IV presents the experimental results and provides relevant analysis, while Section V delivers the conclusion.

II. Related Work

A. Mono-Modal Image Registration

For image registration in the same modality, the only difference between the image pairs to be aligned is spatial position. According to the type of registration, the model can be classified into two categories, one with rigid registration and the other with nonrigid registration.

Rigid registration involves linear transformations including rotation, translation, scaling, and projection. In the traditional feature-based methods, the image pairs first are subjected to feature extraction, including feature point, feature line, and feature region. Some classical methods for feature detection and description include SIFT [29], SURF [30], RIFT [31], the Canny detector [32], and Laplacian of Gaussian-based detectors [33]. Then, the correspondence relationship between images can be established based on the detected feature. Finally, the moving images are resampled using a learned transformation relationship. Sourabh Paul et al. [34] designed an effective matching scheme based on the initial matching pairs obtained by SIFT to increase the number of correct matches. Traditional methods are overly dependent on the quality of feature extraction, too few feature points extracted or a high feature matching error rate can lead to suboptimal registration results. With the development of deep networks, DeTone et al. [35] first applied deep neural networks to the task of homography matrix estimation, and this approach enabled fast and accurate registration. To improve the registration accuracy, Zhang et al. [36] proposed content-aware masks, which can reject outliers when applied to deep features used for registration. This leads to more accurate registration. The learning-based approach significantly reduces run time and enables more accurate registration.

For nonrigid registration, i.e., pixel-by-pixel transformations, traditional methods usually implement the registration process by optimizing the cost function between the pairs of images, such as NCC and MI. However, these methods require continuous iterative optimization, which leads to high computational costs. Later, with the rise of CNNs. Cao et al. [37] introduced a CNN-based regression model for medical image registration, where the model learns deformation fields directly from unaligned image pairs. Balakrishnan et al. [38] proposed an unsupervised framework called Voxelmorph, which utilizes a U-Net [39] architecture to learn the deformation field between unaligned medical images.

B. Multimodal Image Registration

Image pairs have significant nonlinear modality gaps in addition to spatial position differences in multimodal image registration tasks, which poses a substantial challenge compared to the registration of mono-modal image. Moreover, it is challenging to achieve optimal performance if multimodal image registration networks are optimized only through the use of similarity operators, and the ground-truth transformation parameters for the remote sensing image registration are difficult to obtain. To address these challenges, researchers usually artificially distort aligned multimodal image pairs to obtain the ground-truth transformation parameters and use them as supervisory information [40], [41], [42]. Simultaneously, they integrate operators that can quantify image similarity as supplementary constraints. This approach enhances the accuracy and robustness of the registration process. Paul and Pati [43] combine MI and two kinds of improved simultaneous perturbation stochastic approximation (SPSA) optimization to achieve coarse-to-fine registration of high-resolution optical-SAR pairs. Li et al. [40] use a two-stream network for registration and combine MI operator and ground-truth transformation parameters to propose a dual-supervised loss for training registration network. Ye et al. [44] propose an incremental registration network and use NCC operator as supervisory information. Xiao et al. [45] design a two-stage registration pipeline to realize the coarse-to-fine registration process and use MI operator and ground-truth transformation parameters as supervisory information. However, these hand-designed image similarity operators are not generic and cannot be applied to all modalities.

With the emergence of GANs, some researchers explore GAN-based modality translation to reduce the registration difficulty of cross-modal registration by translating it into a mono-modal registration problem. For example, Wang et al. [46] propose a cross-modal perceptual style transformation network based on CycleGAN [47] for performing the registration task. Wei et al. [48] use CycleGAN to perform the translation of MR-CT images, ultimately achieving cross-modal medical image registration. However, GAN-based methods mainly aim to remap the data distribution between different modalities without guaranteeing that the translated image precisely aligns with the modality of the target image, and the image translation may unavoidably introduce artificial noise, which can adversely affect the registration process. Hence, we design a PME and MPM to address these problems.

III. Method

As depicted in Fig. 2, the proposed MMRNet is divided into two stages: training the PME and training the registration network. Next, we will offer a comprehensive description of the overall pipeline and the specific structures of these two components in Sections III-A–III-C.

A. Overall Pipeline

1) Stage 1: Training PME: Traditional image similarity operators are typically designed based on mathematical priors, primarily calculating the correlation between the registered image and the fixed image to estimate the registration accuracy. Since the computation of these operators is

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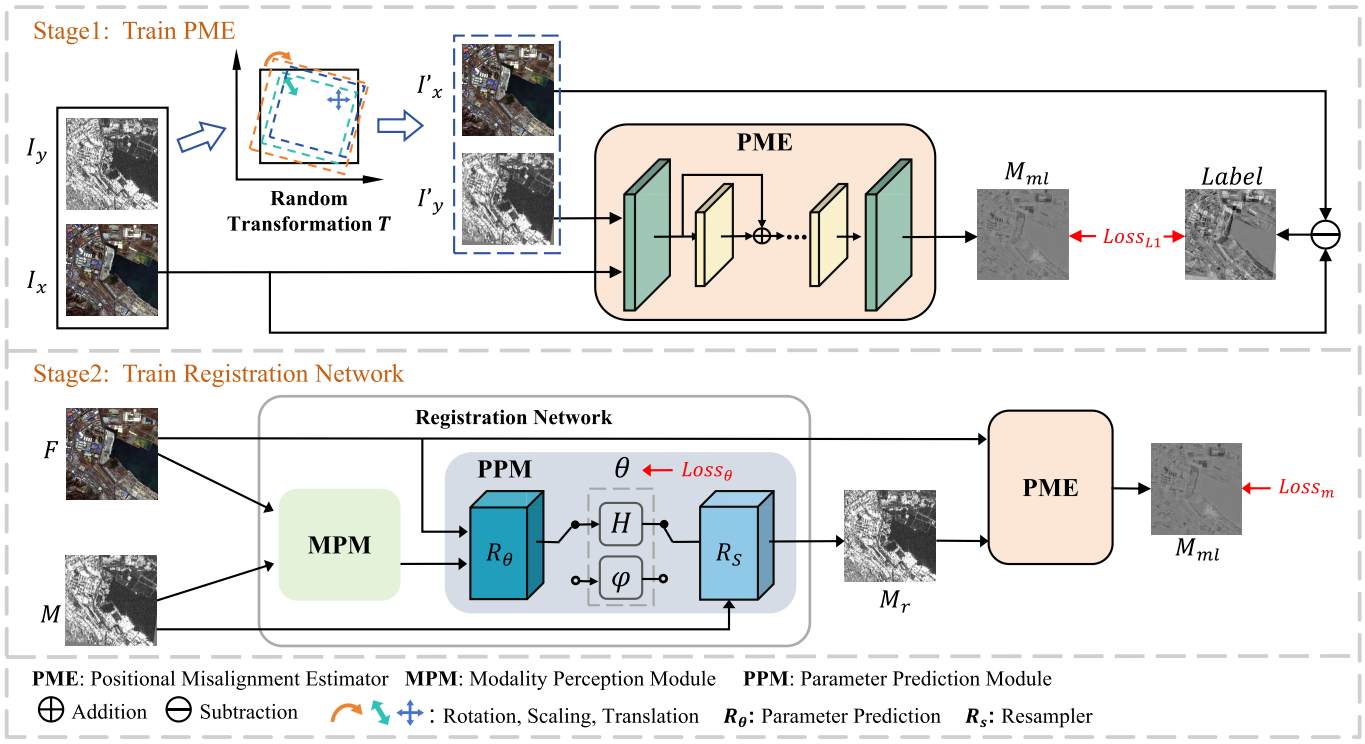


Fig. 2. Overview framework of the proposed method. There are two main stages: training the PME and training the registration network.

data-independent, they generally maintain high accuracy in single-modal image registration tasks. However, when applied to multimodal image registration, these similarity operators often lose their effectiveness due to modality differences. Additionally, traditional similarity operators focus more on the semantic information of the images, whereas image registration tasks prioritize the positional information. To overcome the limitations of traditional similarity operators, inspired by [14], we propose PME.

The role of PME is to estimate the positional differences between multimodal images and quantify the accuracy of the registration. A ground-truth label is required for obtaining an accurate misalignment map. However, obtaining real labels in real scenarios is challenging. Therefore, we employ a self-supervised strategy for training the PME. Specifically, as shown in Fig. 2, we perform random affine transformations T on the image pairs of different modalities (denoted as I_x and I_y) to introduce misalignment in their spatial position, including rotation, translation, and scaling, resulting in I'_x and I'_y, respectively. Then, we subtract I'_x from I_x. Since they are images of the same modality, the only difference between them lies in their spatial positions. This difference can be considered as the ground-truth misalignment map, which we utilize as labels. I'_y and I_x are then fed into the PME, and a predicted misalignment map M_{ml} is output. Our PME uses ResNet [28] as backbone based on its strong feature extraction capability and its proven effectiveness in learning complex representations. And the loss between the predicted misalignment map and ground-truth label is calculated to optimize the performance of the estimator. Similar to work [14], after pretraining, a PME that can overcome the modality gaps and learn the spatial positional differences is obtained. We can represent this process as

$$\left(I'_x I'_y\right) = \text{AT}\left(\left(I_x, I_y\right), T\right) \tag{1}$$

$$\text{Label} = I'_x - I_x \tag{2}$$

$$M_{\text{ml}} = \text{PME}\left(\text{Concat}\left(I'_y, I_x\right)\right) \tag{3}$$

$$\text{Loss}_{L1} = \frac{1}{hw}\sum_{i,j}^{h,w} |M_{\text{ml}}(i,j) - \text{Label}(i,j)|_1 \tag{4}$$

where h and w denote the size of the misalignment map, which is consistent with the size of input. $\text{AT}(\cdot)$ denotes the affine transformation. (i, j) indicates the coordinates in the map. The goal of stage 1 is to minimize Loss_{L1} to enhance the estimation accuracy of the PME.

2) Stage 2: Training Registration Network: The pretrained PME in Stage 1 serves as auxiliary supervisory information to guide the training of the registration network in Stage 2. Specifically, the registration network takes the image pair M (moving image) and F (fixed image) to be aligned as input, producing the registered image M_r [see (5)] (the detailed process can be found in Section III-B). Then, M_r and F are fed into the pretrained PME to obtain the misalignment map M_{ml} [see (6)]. Subsequently, the registration network is optimized by minimizing M_{ml} along with other associated losses (the detailed description can be found in Section III-C). Note that during the training of the registration network, the parameters of the PME are frozen

$$M_r = \text{RegN}(M, F) \tag{5}$$

$$M_{\text{ml}} = \text{PME}(\text{Concat}(M_r, F)) \tag{6}$$

where RegN denotes the registration network. Since PME uses a CNN to estimate the misalignment map between registration image and fixed image, compared to traditional operators,

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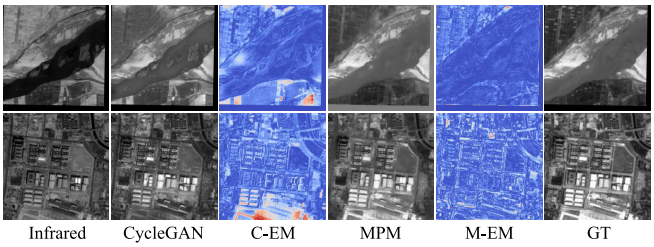


Fig. 3. Infrared represents the source infrared image, the CycleGAN and C-EM represent the modality translation result generated by the CycleGAN and the error map with the target image, and the MPM and MPM error map (M-EM) represent the modality translation result generated by the MPM and the error map with the target image, respectively. GT denotes the target optical image.

it is more flexible and has better generalization ability when handling various types of images and tasks.

B. Registration Network

The overall structure of the registration network is illustrated at the bottom of Fig. 2, comprising two main modules: MPM and parameter prediction module (PPM).

1) Modality Perception Module: The modality translation method based on GAN achieves translation by remapping the distributions. However, the distributions have biases and variances, making it difficult to achieve perfect target-level correspondence. When dealing with data that have significant modality differences, additional noise may be generated. As shown in Fig. 3, although the translated image generated by CycleGAN is visually the same modality as the target image, there are still significant differences in the specific target region. To solve this problem, we designed MPM to achieve high-quality modality translation, and it is simple to operate and implement.

The primary function of MPM is to learn modality differences and perform modality translation, alleviating the significant modality gaps between different modality images in the registration task. The detailed structure and modality translation flow are shown in Fig. 4. The moving image M and fixed image F are first concatenated in the channel dimension and fed into MPM to learn the modality differences D_m between them. Then, we add the learned D_m to the moving image M, resulting in a pseudo-modality image M_t with a similar modality to F. Finally, the pseudo-modality images M_t and F are fed into the PPM to predict the transformation parameters θ between image pairs. The formulation process is as follows:

$$D_m = \text{MPM}(\text{Concat}(M, F)) \tag{7}$$

$$M_t = M + D_m. \tag{8}$$

As we know, registration task within the same modality is relatively easier compared to registration task involving different modalities. Therefore, the effectiveness of the MMR-Net also depends partially on the effectiveness of modality translation, i.e., the effectiveness of MPM. We also use ResNet as the backbone of MPM, and its deep feature extraction ability allows MPM to capture these modality-specific characteristics and enhance the understanding of modality information.

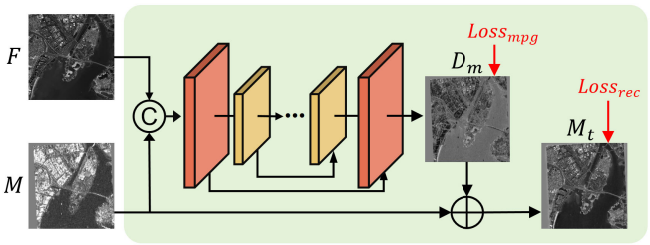


Fig. 4. Overview structure of the MPM. The translated image M_t and the fixed image F are fed into the PPM to predict θ.

Additionally, to ensure the fidelity of the translated images, we introduce a reconstruction loss to constrain the generation of images. The reconstruction loss is presented in Section III-C for details.

2) Parameter Prediction Module: R_θ and R_s are designed in PPM. R_θ aims to predict the transformation parameter θ, and R_s distorts the image based on the predicted transformation parameter θ. We integrate two types of registration models (rigid and nonrigid registrations) in R_θ to verify the generality of our proposed framework, and we give detailed description, respectively, in the following.

a) Rigid registration: For rigid registration, i.e., linear transformation, we use the affine transformation, which is performed by using a 3×3 matrix, denoted as H. The components of this matrix represent the degrees of rotation, translation, and scaling, respectively. The mapping relation can be expressed as follows:

$$\begin{bmatrix} u' \\ v' \\ 1 \end{bmatrix} = \begin{bmatrix} h_0 & h_1 & h_2 \\ h_3 & h_4 & h_5 \\ h_6 & h_7 & h_8 \end{bmatrix} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{9}$$

where u' and v' denote the new coordinates after the transformation, H has six degrees of freedom, and h_6, h_7, and h_8 are constrained to be 0, 0, and 1, respectively. The detailed structure is shown in Fig. 5(a), and PPM consists of two feature extractors [49] and a ResNet [28] architecture module; although the MPM processing greatly reduces the difficulty of the registration, we further extract the more critical information for the registration through feature extractor. F and M_t are inputs, and the affine transformation matrix H can be obtained. We feed moving image M and affine matrix H to R_{s_H} to obtain the registration image M_r. The formulas are expressed as follows:

$$H = R_{\theta_H}(M_t, F) \tag{10}$$

$$M_r = R_{s_H}(M, H) \tag{11}$$

where R_{θ_H} and R_{s_H} denote R_θ and R_s for rigid registration, respectively, and R_{s_H} warps the moving image M according to (9) for registration.

b) Nonrigid registration: For nonrigid registration, the primary objective is to estimate the deformable field φ with the same size as the input image. This field has two channels representing the displacement of each pixel along the x- and y-axes, respectively. The deformable field φ is then applied to M to achieve pixelwise displacement, enabling the registration process. As in rigid registration, the inputs are M_t and F and

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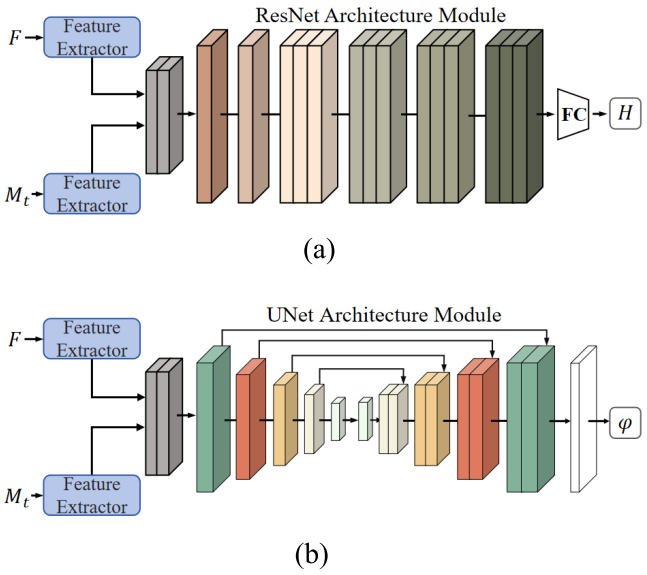


Fig. 5. Structure of the two types of PPM integrated in R_θ. (a) Affine estimation network for rigid registration. (b) Deformable field estimation network for nonrigid registration.

the output is φ. The formulas are expressed as follows:

$$\varphi = R_{\theta_\varphi}(M_t, F) \tag{12}$$

$$M_r = R_{s_\varphi}(M, \varphi) \tag{13}$$

where R_{θ_φ} and R_{s_φ} denote R_θ and R_s for nonrigid registration, respectively, and R_{s_φ} distorts M according to φ to obtain the registration image M_r. As shown in Fig. 5(b), we similarly use two feature extractors and employ U-Net [39] as the backbone for deformable field estimation.

The detailed description of MMRNet can be seen in Algorithm 1.

Algorithm 1 Algorithm of MMRNet

Require: Fixed images $\{I_x\}$; Moving images $\{I_y\}$; Random Affine Transformation AT; Positional Misalignment Estimator PME;
Ensure : A trained registration network MMRNet

Stage 1: Train the PME
while *not converged* **do**
  Sample a mini-batch from $\{I_x\}$ and $\{I_y\}$;
  Obtain (I'_x, I'_y) according to Eq. (1);
  Obtain $Label$, M_{ml} according to Eq. (2) and Eq. (3);
  Compute the loss between $Label$ and M_{ml} according to Eq. (4);
  Update PME according to Eq. (4);
end
Stage 2: Train the Registration Network
while *not converged* **do**
  Sample a mini-batch from $\{I_x\}$ and $\{I_y\}$;
  Obtain (F, M) according to Eq. (1);
  Obtain M_t according to Eq. (7) and (8);
  Predict H/φ according to Eq. (10) or Eq. (12);
  Resample M according to Eq. (11) or Eq. (13);
  /* H denotes rigid, φ denotes nonrigid */
  Update Registration Network according to Eq. (21);
end

C. Loss Functions

Compared with [14], we design a more comprehensive combination of loss functions. To train the registration network in stage 2, the total loss consists of parameter loss L_θ, misalignment loss L_m, reconstruction loss L_{rec}, and modality perception guidance loss L_{mpg}. For the nonrigid registration, we additionally add a smooth loss L_s to ensure that the deformable field does not distort rapidly. The details of these losses are introduced as follows.

1) Parameter Loss: L_θ is used to constrain the gap between the ground-truth transformation parameters θ_{gt} and the predicted transformation parameters θ_{pre}. The formula is expressed as follows:

$$L_\theta = \left\|\theta_{\text{pre}} - \theta_{\text{gt}}\right\|_1 \tag{14}$$

for rigid registration, θ denotes H, and for nonrigid registration, θ denotes φ. The nonrigid ground-truth transformation parameters are obtained by encoding the rigid ground-truth transformation parameters.

2) Misalignment Loss: The registration image M_r and the fixed image F are fed into the PME to obtain the misalignment map M_{ml}. If the two images are aligned, the M_{ml} between them should be zero, so we optimize the registration network by minimizing M_{ml} as

$$L_m = \text{AVG}(|M_{\text{ml}}|) \tag{15}$$

where $|\cdot|$ indicates an absolute value, and AVG$(\cdot)$ denotes the operation of calculating the average value.

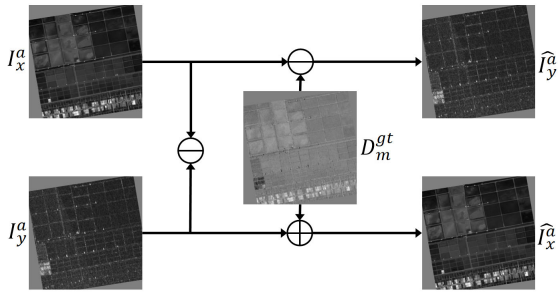


Fig. 6. Relationship between modality differences and aligned multimodal image pairs.

3) MPM Loss: There are two kinds of losses involved in optimizing MPM, including modality perception guidance loss L_{mpg} and reconstruction loss L_{rec}. We perform a subtraction operation on the aligned images of different modalities to obtain the ground-truth modal differences D_m^{gt}. They can be translated into each other by adding or subtracting D_m^{gt}. The relationship is shown in Fig. 6, and the formula is expressed as follows:

$$\begin{aligned} \hat{I}_y^a &= I_x^a - D_m^{\text{gt}} \\ \hat{I}_x^a &= I_y^a + D_m^{\text{gt}}. \end{aligned} \tag{16}$$

We use D_m^{gt} to impose constraints on the MPM. These two kinds of losses optimize the modality translation from different

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TABLE I
COMPARISON OF COMMON MULTIMODAL REMOTE SENSING REGISTRATION DATASETS

Dataset	Modality Info.	Size	Resolution(m)	Quantity	Alignment	Purpose
SARptical [51]	optical-SAR	112*112	optical:0.2, SAR:1	10108	No	data analysis
SEN 1-2 [52]	optical-SAR	256*256	optical:10, SAR:10	28384	Yes	data fusion
OSdataset [50]	optical-SAR	256*256	optical:1, SAR:1	10692	Yes	image registration
QXS-SAROPT [53]	optical-SAR	256*256	optical:1, SAR:1	20000	No	data fusion
GoogleMap [17]	optical-map	192*192	-	9710	Yes	homography estimation
MRSR(Ours)	optical-SAR	256*256	optical:10, SAR:10	3850	Yes	image registration
	optical-infrared	256*256	optical:10, infrared:10	4000	Yes	image registration

perspectives, and the sum of them is denoted as L_{mpm}, i.e.,

$$L_{\text{mpg}} = \left\| D_m - D_m^{\text{gt}} \right\|_2^2 \tag{17}$$

$$L_{\text{rec}} = \left\| F_{\text{gt}} - M_t \right\|_2^2 \tag{18}$$

$$L_{\text{mpm}} = L_{\text{rec}} + L_{\text{mpg}} \tag{19}$$

where F_{gt} denotes the image obtained by F with the same spatial transformation as M.

4) Smooth Loss: In the nonrigid registration task, to enhance the smoothness and continuity of the predicted deformable field and avoid unnatural transformations, we introduce a smooth loss as follows:

$$L_s = \frac{1}{N} \sum_{p \in \varphi} \left[\left(\frac{\partial \varphi(p)}{\partial x} \right)^2 + \left(\frac{\partial \varphi(p)}{\partial y} \right)^2 \right] \tag{20}$$

where p denotes the element in the deformable field, and N denotes the total number of elements.

The total loss for training in Stage 2 can be calculated as follows:

$$\text{Loss} = \lambda_\theta L_\theta + \lambda_{\text{mpm}} L_{\text{mpm}} + \lambda_m L_m + \lambda_s L_s. \tag{21}$$

The weights of each loss term are denoted by λ_θ, λ_{mpm}, λ_m, and λ_s. For specific weight analysis and selection, see Section IV-D. Note that the smooth loss weight λ_s is set to 0 when we perform rigid registration.

IV. EXPERIMENTS

A. Datasets and Implementation Details

1) Analysis of the Existing Datasets: Due to the lack of publicly available datasets, learning-based remote sensing image registration methods struggle to achieve well-generalized network models. Although several commonly used datasets exist in this field, they typically contain only a single type of multimodal data and often exhibit low alignment accuracy. This limitation is mainly because some datasets are originally designed for other tasks, such as data fusion. Consequently, registration models trained on these datasets may not generalize well to real-world multimodal remote sensing image registration tasks. As shown in Table I, we summarize several commonly used datasets. For optical-SAR data, only the OSdataset [50] is specifically designed for the registration task. The SARptical dataset [51] has a small image size of only 112 × 112, and its optical-SAR image pairs are not strictly aligned. The SEN1-2 [52] and QXS-SAROPT [53] datasets were originally introduced for data fusion tasks, and the QXS-SAROPT dataset also lacks precisely aligned image pairs. While the GoogleMap [17] dataset, designed for homography estimation, provides accurately aligned image pairs, its original image size is only 192 × 192, which is relatively small. There are no publicly available optical-infrared remote sensing registration datasets. Therefore, developing a comprehensive multimodal remote sensing image registration dataset is crucial for advancing this field.

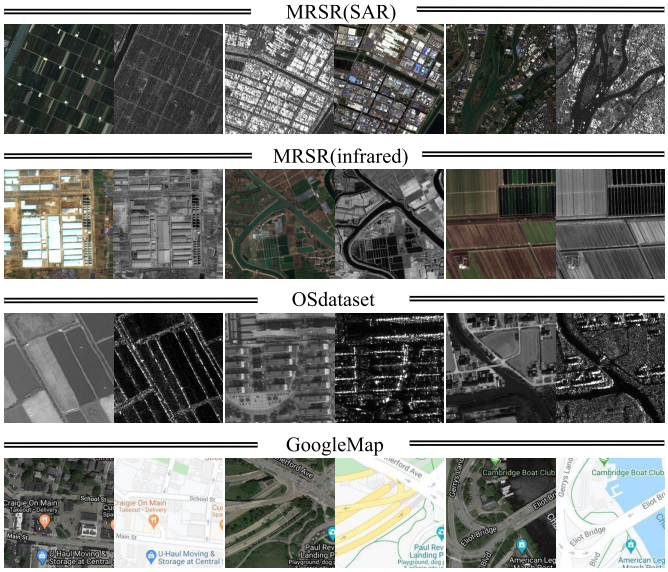


Fig. 7. Examples of optical-SAR and optical-infrared data in MRSR dataset, OSdataset, and GoogleMap dataset.

2) Experimental Datasets: We have constructed a new registration dataset named the MRSR dataset, primarily based on Sentinel-1 and Sentinel-2 data [54]. It includes 3850 optical-SAR image pairs and 4000 optical-infrared image pairs, with each image sized at 256 × 256 pixels and a spatial resolution of 10 m. The dataset encompasses a wide variety of scenes such as urban regions, agricultural land, rivers, and mountains. To further assess the generalization and robustness of our method, we also perform evaluations on the OSdataset [50] and GoogleMap dataset [17]. The OSdataset features high-resolution, co-registered optical and SAR patch pairs, with SAR images from Gaofen-3 and optical images from Google Earth, all at 1-m spatial resolution. A total of 4000 image pairs are randomly sampled for testing. The GoogleMap

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TABLE II
RIGID REGISTRATION RESULTS OF VARIOUS METHODS. RMSE DENOTES THE ROOT-MEAN-SQUARE ERROR OF THE CORNER POINT DISPLACEMENT. THE BEST RESULTS ARE MARKED IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED

Methods	SAR→optical			infrared→optical			OSdataset			GoogleMap		
	RE ↓	$RMSE_c$ ↓	NCC ↑	RE ↓	$RMSE_c$ ↓	NCC ↑	RE ↓	$RMSE_c$ ↓	NCC ↑	RE ↓	$RMSE_c$ ↓	NCC ↑
ORB [57]	45.831	212.588	0.0843	56.056	215.796	0.0913	55.992	214.151	0.0843	72.948	239.015	0.0876
RIFT [31]	23.458	21.355	0.1681	32.608	28.256	0.2539	33.003	47.466	0.0930	22.736	13.298	0.2537
TWMM [58]	15.959	7.257	0.4427	22.331	7.549	0.5036	29.323	18.374	0.1191	10.358	4.064	0.4052
IHN [59]	5.060	0.909	0.7857	4.765	0.975	0.7779	5.614	1.031	0.7803	3.282	0.915	0.6557
InMIRNet [27]	13.491	4.905	0.5436	9.698	1.810	0.5782	17.403	2.343	0.3731	7.332	2.605	0.4658
DCNet [60]	11.096	2.596	0.5359	14.497	2.022	0.4137	14.507	2.229	0.4014	5.688	1.438	0.6018
RHWF [61]	5.385	0.955	0.7740	5.039	1.055	0.7710	7.464	1.633	0.7243	5.122	1.596	0.5998
MCNet [62]	8.053	1.388	0.6878	11.993	2.614	0.4791	8.769	1.551	0.6627	2.569	0.713	0.6783
Ours	**4.511**	**0.858**	**0.7941**	**3.909**	**0.716**	**0.8513**	**5.523**	**0.888**	**0.7922**	**1.401**	**0.475**	**0.7336**

dataset, constructed via the Google Maps Static API, contains multiband satellite images paired with corresponding Google Maps visuals. Similarly, we randomly selected 4000 pairs for experiments. Fig. 7 shows some sample images.

To synthesize unaligned images for training the network, as in [14], we apply random affine transformations to moving images with $[-10°, 10°]$ rotation, $[-10\%, +10\%]$ displacement, and $[-12\%, +12\%]$ scaling. All datasets are selected with 3500 pairs as training set, with the remaining pairs allocated to the test set.

3) Implementation Details: The environment for our experiments is PyTorch 1.9 [55], Intel Core i7-12700F 2.10-GHz CPU, and NVIDIA RTX 3090 GPU. The Adam [56] optimizer is used to train our model, employing a batch size of 8 and a learning rate of 0.0001. The hyperparameters are carefully chosen as follows: $\lambda_\theta = 10$, $\lambda_{\text{mpm}} = 2.5$, $\lambda_m = 5$, and $\lambda_s = 1$.

B. Evaluation Metrics

Three common metrics are used to evaluate registration performance, including reprojection error (RE) [41], root-mean-square error (RMSE) [35], and NCC. RE serves as a primary metric for assessing the accuracy of registration and is used to quantify the error between the ground-truth registration result and the predicted registration result. We use L1 loss [59] for this calculation. NCC calculates the similarity between the aligned image and the ground-truth-aligned image to evaluate the registration accuracy. RMSE of the corner point displacements is utilized for rigid registration (denoted as RMSE_c). For nonrigid registration, RMSE is calculated between the predicted deformable field and the ground-truth deformable field (denoted as RMSE_d). The formulas are expressed as follows:

$$\text{RE} = \frac{1}{n}\sum_{i=1}^{n}\left|M_{r_i} - M_{r_i}^{\text{gt}}\right| \tag{22}$$

where M_{r_i} denotes the registration result predicted by the model, $M_{r_i}^{\text{gt}}$ denotes the corresponding ground truth, and n represents the number of samples

$$\text{RMSE}_c = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{4}\sum_{j=1}^{4}\sqrt{\left(x_j^{\text{pre}} - x_j^{\text{gt}}\right)^2 + \left(y_j^{\text{pre}} - y_j^{\text{gt}}\right)^2} \tag{23}$$

where $(x_j^{\text{pre}}, y_j^{\text{pre}})$ and $(x_j^{\text{gt}}, y_j^{\text{gt}})$ denote the coordinates of the four corner points calculated from the predicted transformation parameters and the ground-truth transformation parameters, respectively,

$$\text{RMSE}_d = \frac{1}{n}\sum_{i=1}^{n}\sqrt{\frac{1}{hw}\sum_{k=1,v=1}^{h,w}\left(\varphi_{k,v}^{\text{pre}} - \varphi_{k,v}^{\text{gt}}\right)^2} \tag{24}$$

where h and w represent the height and width of the image, respectively. $\varphi_{k,v}^{\text{pre}}$ represents the components in the predicted displacement vector field. $\varphi_{k,v}^{\text{gt}}$ represents the corresponding ground-truth value.

C. Results and Analysis

In this article, we compare our method with some traditional and recent approaches of rigid registration: ORB [57], TWMM [58], RIFT [31], InMIRNet [27], DCNet [60], IHN [59], MCNet [62], and RHWF [61]. We also compare our method with the following nonrigid registration methods: UMFusion [46], TransMorph [63], SuperFusion [41], ADRNet [45], CHANet [4], NBRNet [64], and InMIRNet [27]. To ensure a fair comparison, all methods in the experiment are retrained and we employ the same test set for evaluation.

1) Quantitative Evaluation: The quantitative evaluation of rigid registration is shown in Table II, we can note that the traditional methods, ORB and RIFT, lose the ability to register remote sensing data with large modality gaps. TWMM and InMIRNet have limited registration capability and cannot achieve high accuracy. DCNet, MCNet, IHN, and RHWF all use iterative strategies to achieve a progressive registration process, which has certain advantages compared with other methods and can achieve accuracy with RMSE_c less than 3. Specifically, the RMSE_c of IHN on all four datasets is approximately 1, and the RMSE_c of RHWF on all four datasets is less than 2. Our approach performs best on four datasets,

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TABLE III

NONRIGID REGISTRATION RESULTS OF VARIOUS METHODS. RMSE DENOTES THE ROOT-MEAN-SQUARE ERROR OF THE DEFORMABLE FIELD. THE BEST RESULTS ARE MARKED IN BOLD, AND THE SECOND BEST RESULTS ARE UNDERLINED

Methods	SAR→optical			infrared→optical			OSdataset			GoogleMap		
	RE ↓	$RMSE_d$ ↓	NCC ↑	RE ↓	$RMSE_d$ ↓	NCC ↑	RE ↓	$RMSE_d$ ↓	NCC ↑	RE ↓	$RMSE_d$ ↓	NCC ↑
UMFusion [46]	36.181	0.1155	0.0961	17.183	0.0599	0.3746	33.271	0.1134	0.0943	10.362	0.0619	0.4031
TransMorph [63]	37.241	0.1324	0.0966	25.869	0.1389	0.1561	35.366	0.1286	0.0904	32.298	0.1314	0.1025
SuperFusion [41]	33.059	0.0437	0.1475	4.177	0.0083	0.8306	9.460	0.0134	0.6626	6.279	0.0121	0.5424
InMIRNet [27]	28.263	0.0987	0.1381	30.469	0.1274	0.1348	26.841	0.0807	0.1192	7.861	0.0176	0.4762
NBRNet [64]	18.851	0.0326	0.3314	9.508	0.0106	0.5806	21.421	0.0311	0.2288	8.956	0.0144	0.4133
CHANet [4]	28.278	0.0535	0.2285	18.612	0.0299	0.3928	26.920	0.0668	0.1603	22.305	0.0335	0.2686
ADRNet [45]	12.887	0.0306	0.4135	5.569	0.0087	0.7836	12.705	0.0141	0.5243	5.679	0.0105	0.6014
Ours	**4.415**	**0.0044**	**0.8011**	**2.107**	**0.0040**	**0.8841**	**6.442**	**0.0047**	**0.9623**	**1.883**	**0.0031**	**0.7676**

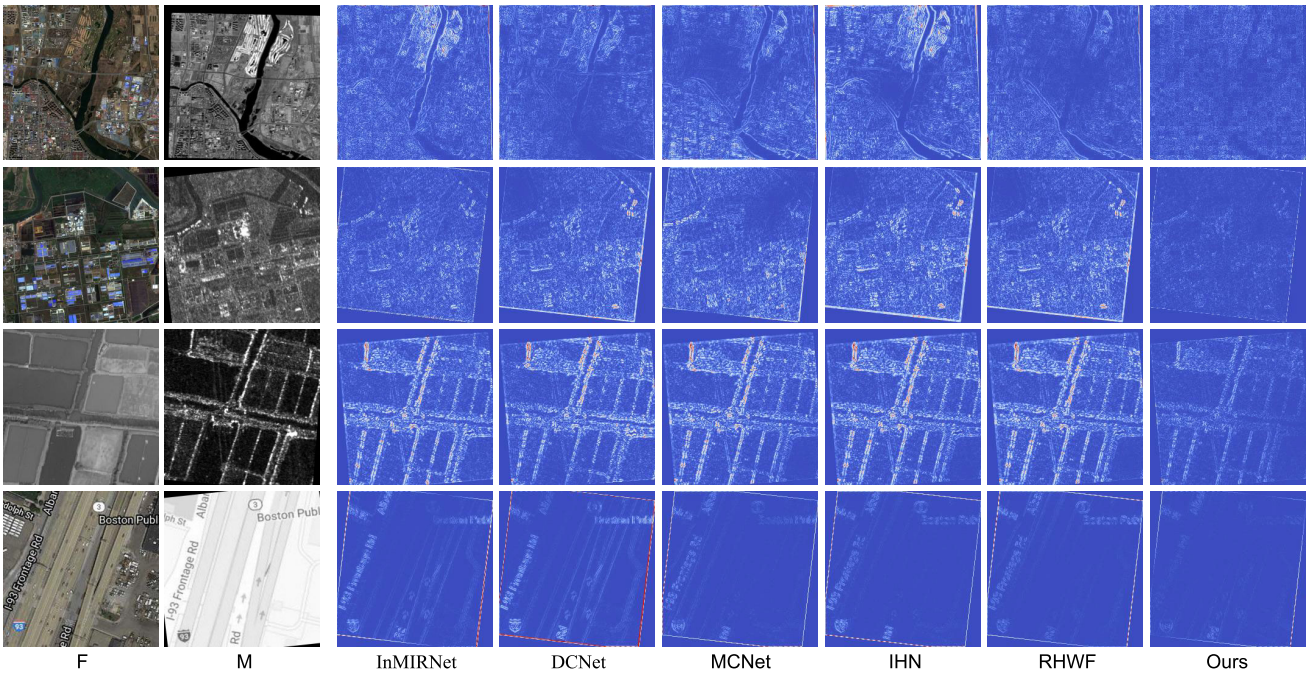


Fig. 8. Example of rigid registration. (From top to bottom) Optical-infrared, optical-SAR, OSdataset, and GoogleMap. F: fixed image and M: moving image.

all with RMSE_c less than 1. Most notably on the GoogleMap dataset with an RMSE_c of 0.475, far superior to the other methods. On the OSdataset, our method is also the only one with RMSE_c less than 1.

The quantitative results of nonrigid registration are listed in Table III. It can be seen that on optical-SAR dataset, many methods have lost their comparability due to their difficulty in learning pixel correspondences influenced by speckle noise in SAR images. ADRNet, CHANet, and NBRNet mitigate this difficulty by using a coarse-to-fine strategy but still fail to achieve high accuracy. Our method can achieve smaller deformable field errors. Similarly, on the optical-infrared dataset with relatively small modality gaps, although some methods such as SuperFusion, NBRNet, and ADRNet show some effectiveness, our method still shows a great advantage. On the GoogleMap dataset with rich structural information, most of the methods have good registration ability. However, our method is still the most performing, achieving RE of 1.883. These results from these sets of experiments verify the effectiveness and superiority of our proposed method. This is attributed to the fact that our trained PME can accurately estimate the spatial positional differences of the image pairs, even if there are significant modality gaps.

2) Qualitative Evaluation: To perform qualitative comparisons, we randomly selected several pairs of images in the test set for each type of data. Figs. 8 and 9 visualize the rigid and nonrigid registration results on the four datasets, respectively. In the rigid registration task, due to the characteristics of the rigid transformation, a little difference in the transformation parameters will lead to the whole image mismatch. Our method uses PME to constrain spatial position differences, which can guide the learning of transformation parameters well. Similarly, in nonrigid registration tasks, our method can also achieve minimal local differences. While other methods are affected by modal differences, it is difficult to correctly learn the direct correspondence between pixels. Our method

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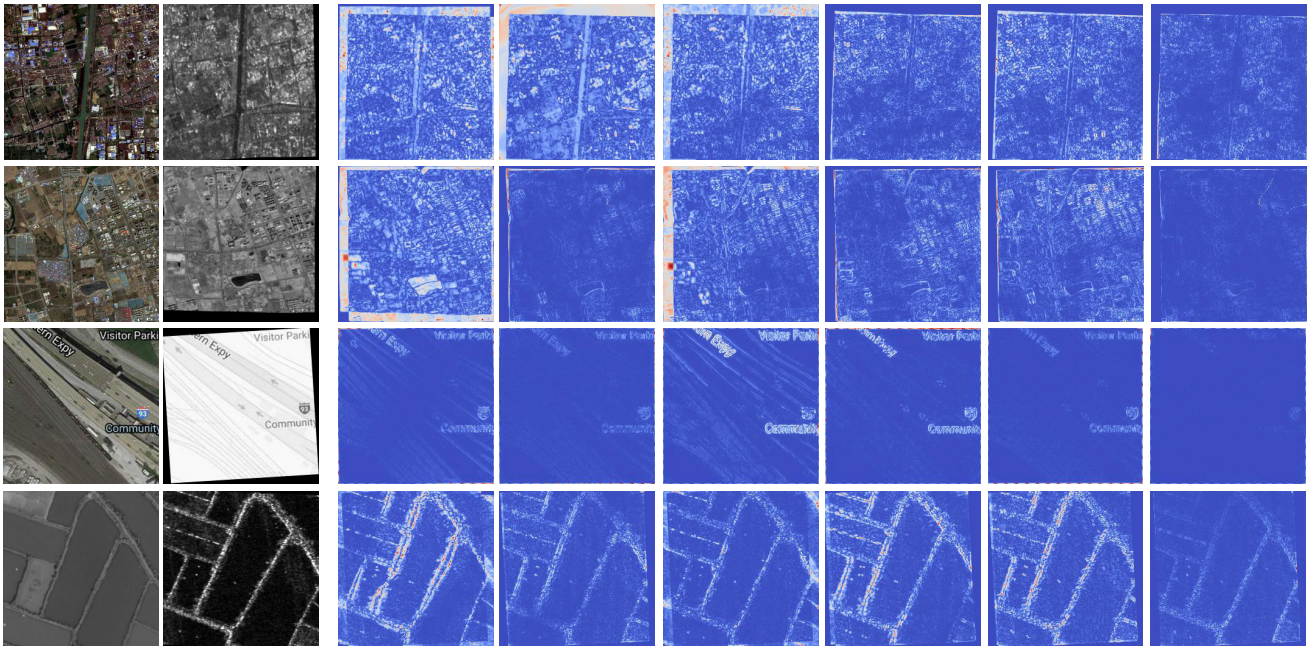


Fig. 9. Example of nonrigid registration. (From top to bottom) Optical-SAR, optical-infrared, GoogleMap, and OSdataset. F: fixed image and M: moving image.

can alleviate the influence of modality gaps through spatial position constraint and high-quality modality translation. The illustrations clearly demonstrate the better performance of our method on various data.

3) Computation Complexity: To comprehensively evaluate model performance, we compared their computational complexity, including the number of parameters and inference time. For the learning-based methods, runtime tests were performed using the same GTX 3090 GPU and Intel Core i7-12700F 2.10-GHz CPU. Detailed information is provided in Table IV. Our method has advantages over most learning-based methods. Compared to InMIRNet$_R$ and SuperFusion, we have very little running time although we have far more parameters than them. Our rigid and nonrigid registration model parameters are different; this is because the PPM is different for rigid and nonrigid registration tasks.

TABLE IV

COMPUTATIONAL EFFICIENCY COMPARISON WITH OTHER METHODS. {Net}$_N$ INDICATES THAT THE MODEL PERFORMS RIGID REGISTRATION. {Net}$_R$ INDICATES THAT THE MODEL PERFORMS NONRIGID REGISTRATION

Methods	Param.(M)	Time(s)
ORB [57]	-	0.060
RIFT [31]	-	0.164
TWMM [58]	-	12.639
UMFusion [46]	0.80	0.019
TransMorph [63]	30.99	0.028
SuperFusion [41]	1.96	0.051
IHN [59]	1.71	0.060
NBRNet [64]	-	0.381
InMIRNet$_R$ [27]	135.04	0.042
DCNet [60]	73.22	0.034
RHWF [61]	1.29	0.157
CHANet [4]	7.14	0.036
InMIRNet$_N$ [27]	0.14	0.021
MCNet [62]	0.85	0.033
ADRNet [45]	75.51	0.058
Ours$_R$	32.65	0.027
Ours$_N$	28.71	0.026

D. Ablation Study

1) Ablation Study on Modules: In order to validate the effectiveness of the individual components proposed in our framework, we perform ablation experiments on two types of registration tasks on optical-SAR data. As we can see from Table V, in the presence of large modality gaps, the addition of supervisory information using PME improves the registration performance. Additionally, it can also be seen that when we use MPM alone, the effectiveness of the registration is optimized as the modality differences are reduced, and the accuracy of the registration is further improved when MPM and L_{mpg} are combined. Finally, our method achieves the best registration results on both types of tasks when the three modules are used simultaneously.

2) Ablation Study on Weights: In the training registration stage, the total loss function is composed of three components: the parameter loss L_θ, the misalignment loss L_m, and the MPM loss L_{mpm}. The total loss is the weighted sum of the three losses, with the weights λ_θ, λ_{mpm}, and λ_m, respectively.

The L_θ is supervised by the ground-truth label and is directly related to the registration parameters, so assigning it a large weight is beneficial for model convergence. The L_{mpm} guides modality translation, and obtaining high-quality

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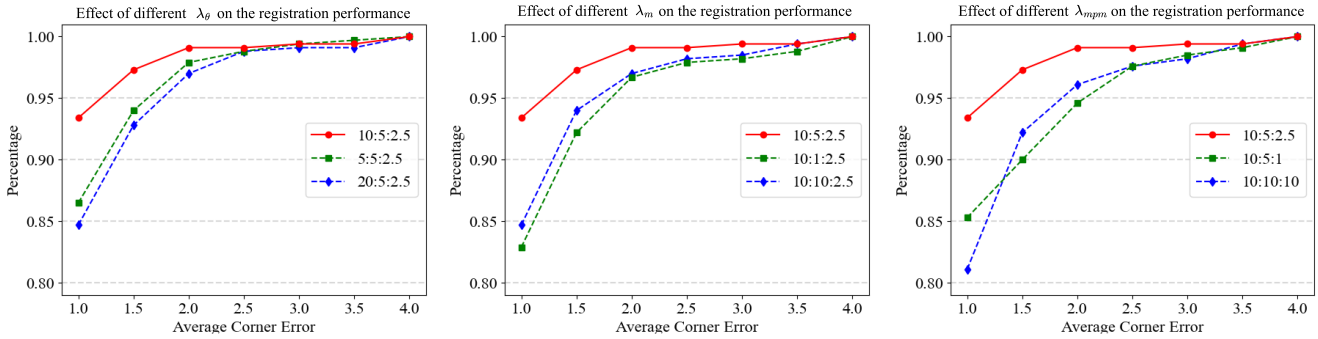


Fig. 10. Ablation experiments are conducted to evaluate the impact of the loss function weights λ_θ, λ_{mpm}, and λ_m. The ratios among these weights are denoted in the legend as λ_θ:λ_m:λ_{mpm}. The x-axis represents the average pixel error (RMSE$_c$), while the y-axis shows the cumulative proportion of test images whose RMSE$_c$ falls below the corresponding x-value. A higher curve reflects improved alignment accuracy.

TABLE V

ABLATION STUDIES OF THE PME, MPM, AND L_{mpg} IN THE OPTICAL-SAR REGISTRATION TASK

Module	PME	MPM	L_{mpg}	Rigid		Non-rigid	
				$RMSE_c\downarrow$	RE$\downarrow$	$RMSE_d\downarrow$	RE$\downarrow$
1	✗	✗	✗	1.066	7.128	0.0069	6.158
2	✓	✗	✗	0.954	6.876	0.0055	5.974
3	✗	✓	✗	0.928	6.342	0.0059	5.433
4	✗	✓	✓	0.876	5.986	0.0046	4.604
5	✓	✓	✓	**0.858**	**4.511**	**0.0044**	**4.415**

TABLE VI

COMPARISON OF THE PERFORMANCE OF REGISTRATION ON THE OSDATASET IN 512 × 512 SIZE

Method	$RMSE_c\downarrow$	RE$\downarrow$	NCC$\uparrow$
IHN [59]	4.538	**13.598**	0.9424
MCNet [62]	4.873	14.531	0.9358
Ours	**3.998**	13.862	**0.9435**

translation result is key to reducing the difficulty of registration. Therefore, the L_{mpm} is also crucial. The L_m constrains spatial position information, ensuring minimal spatial position differences between images, and is used for fine-tuning during coarse registration.

To examine the influence of various weight combinations on registration performance, we compare the test results of models trained under different weight ratios. As shown in Fig. 10, we can see that our method reaches its best when the weights λ_θ:λ_m:λ_{mpm} are 10:5:2.5. Higher or lower weights result in suboptimal registration performance.

E. Additional Analysis

1) Experiments With Larger Images: To demonstrate the registration performance of our method on larger data sizes, we experimented with OSdataset because it provides optical-SAR image pairs in 512 × 512 size. The results are shown in Table VI, and we can see that our approach still shows advantages. Our method achieves RMSE$_c$ less than 4 pixels.

TABLE VII

PME TRAINED WITH DATA GENERATED BY NONRIGID TRANSFORMATIONS AND EXPERIMENTED ON THE OPTICAL-SAR REGISTRATION TASK

Types	Methods	optical-SAR	
		RE$\downarrow$	$RMSE_d\downarrow$
Non-rigid	InMIRNet [27]	28.089	0.1015
	SuperFusion [41]	34.307	0.0446
	ADRNet [45]	14.318	0.0351
	Ours	**4.476**	**0.0043**

TABLE VIII

COMPARISON OF REGISTRATION RESULTS FOR SEVERAL TRADITIONAL ALGORITHMS AND CYCLEGAN IN THE OPTICAL-SAR REGISTRATION TASK

Methods	Rigid		Non-rigid	
	$RMSE_c\downarrow$	RE$\downarrow$	$RMSE_d\downarrow$	RE$\downarrow$
NCC [7]	1.306	9.559	0.1352	35.584
MI [10]	15.025	21.116	0.3837	34.419
MIND [13]	0.928	6.199	0.0278	14.981
CycleGAN [47]	1.270	8.453	0.0059	5.545
Ours	**0.858**	**4.511**	**0.0044**	**4.415**

2) PME for Nonrigid Position Difference Estimation: In the previous experiments, we only used affine transformation to process the data when training the PME. In order to verify the generalization of the PME, on the basis of the affine transformation, we add elastic transformation and Gaussian smoothing to generate nonrigid transformations with a deformation degree of 130 and a Gaussian smoothing radius of 20. The training process of PME remains the same. As shown in Table VII, we compare the three better performing methods: InMIRNet, SuperFusion, and ADRNet. It can be seen that our PME has applicability in nonrigid position differences estimation as well. Meanwhile, the effect of ADRNet is decreased, and this is because ADRNet is a two-stage model, where the first stage is to perform affine transform coarse registration, so when the model processes the image pair with elastic transformation, the coarse registration effect of the first stage will be reduced or invalid.

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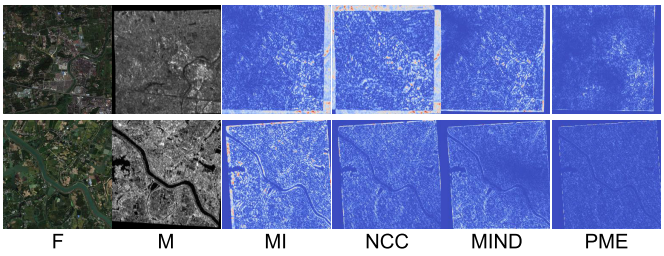


Fig. 11. Visualization of different supervisory information. F: fixed image and M: moving image.

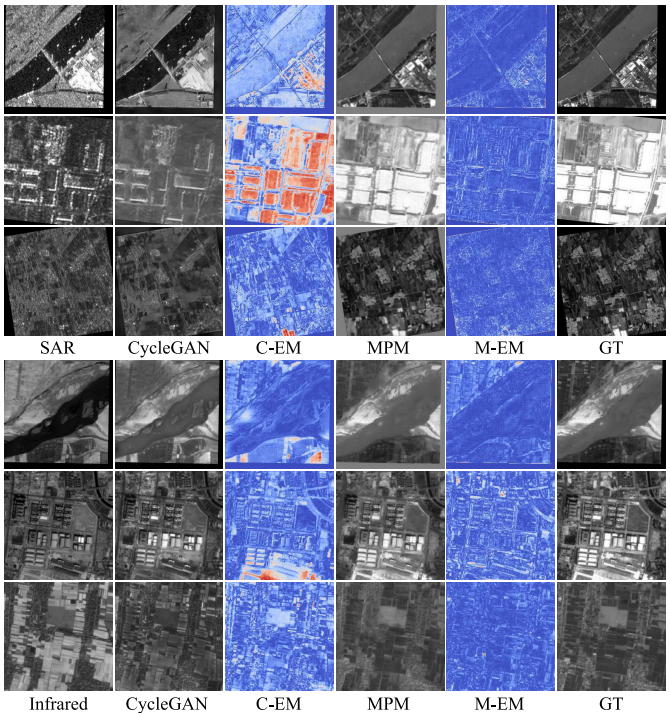


Fig. 12. First column represents the source infrared image, the CycleGAN and C-EM represent the modality translation result generated by the CycleGAN and the error map with the target image, and the MPM and M-EM represent the modality translation result generated by the MPM and the error map with the target image, respectively. GT denotes the target optical image.

3) PME Versus Traditional Operators: To demonstrate the advantages of PME, we conduct a comparative analysis with several traditional similarity operators, including NCC [7], MI [10], and MIND [13]. We still use optical-SAR data for evaluation, and the experimental results are shown in Table VIII. We can note that although the modality gaps between images have been reduced by MPM processing, the results of traditional methods are still not satisfactory. Particularly, the performance of NCC and MI is no longer comparable in nonrigid task. Although MIND exhibits some advantages over NCC and MI, it still falls short when compare to our proposed PME method. We visualized the registration results of supervised information with different image similarity operators, as shown in Fig. 11. Under the guidance of PME, our model obtained the best results. These observations demonstrate the capability of PME to effectively overcome the modality gaps to accurately estimate the positional misalignment.

4) MPM Versus CycleGAN: The MPM assumes a crucial role in executing modality translation, thus easing the registration task. In this section, we compare it with CycleGAN [47], employed as an alternative to MPM. We present quantitative evaluations of the optical-SAR registration in Table VIII. Remarkably, our method surpasses the results obtained via CycleGAN on both types of registration tasks. The visualization results after modality translation are shown in Fig. 12. Noteworthy, our results obtain higher similarity to the ground-truth images in comparison to the outcomes derived from the utilization of CycleGAN.

V. Conclusion

In this study, we propose a novel framework for the registration of multimodal remote sensing image, called MMRNet. First, to explore effective supervisory information, we design a PME to measure the degree of positional differences between image pairs and optimize the registration by minimizing the error. Simultaneously, to mitigate the impact of modality gaps, we develop a new paradigm for modality translation, termed MPM. This is achieved by directly learning modality differences even in the presence of positional misalignment. To ensure high-quality execution of modality translation, we incorporate a modality perception guidance loss for supervised learning of modality differences. Finally, we build a new MRSR dataset for multimodal remote sensing image registration. The extensive experimental results demonstrate the superiority of our method on multimodal remote sensing image registration task.

Despite the achievements made in this study, there are still some unresolved challenges in multimodal remote sensing image registration, including effective supervisory information, general models, and publicly available datasets. Although the proposed PME effectively estimates positional misalignment, it still requires retraining on different datasets, which could lead to efficiency issues in practical applications. In the future, we aim to develop a more general deep learning-based method that can adapt the module to different types of remote sensing data and scenarios, reducing the need for repeated training. Second, our current framework still requires selecting one of the two transformation models for registration. Combining these two transformation models effectively and processing them simultaneously within the same framework remains an open problem worth exploring.

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